

AMENDMENT NO. 1 APRIL 2011
TO
IS 4446 (PART 1) : 2008 CHLOROPHYLL, FOOD
GRADE — SPECIFICATION

PART 1 MAGNESIUM COMPLEX

(Second Revision)

[Page 1, Table 1, *Sl No. (ii), col 2*] — Substitute 'Residual solvents (Acetone, methanol, ethanol, propan-2-ol, hexane), mg/kg, *Max*' for 'Residual solvents, mg/kg, *Max* Acetone, methanol, ethanol, propan-2-ol, hexane'" and delete footnote.

[Page 1, Table 1, *Sl No. (ii), col 4*] — Substitute 'B' for '—'.

[Page 1, Table 1, *Sl No. (iii), col 2*] — Substitute 'Dichloromethane, mg/kg, *Max*' for 'Dichloromethane'" and delete footnote.

[Page 1, Table 1, *Sl No. (iii), col 4*] — Substitute 'B' for '—'.

[Page 2, clause 5.3(d)] — Substitute 'Net quantity' for 'Net content'.

(Page 3, Annex A) — Insert the following new Annex after Annex A:

ANNEX B

[Table 1, *Sl No. (ii) and (iii)*]

DETERMINATION OF RESIDUAL SOLVENTS AND
DICHLOROMETHANE USING HEAD SPACE CHROMATOGRAPHY

B-1 METHOD 1 (DETERMINATION CARRIED OUT IN WATER)

B-1.1 Reagents

B-1.1.1 *Blank*, sample with very low solvent content.

B-1.1.2 *Internal Standard*, 3-methyl-2-pentanone.

B-1.1.3 *Methanol*

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B-1.1.4 Demineralized Water

B-1.1.5 Internal Standard Solution — Add 50.0 ml water to a 50 ml injection vial and seal. Accurately weigh and inject 15 µl 3-methyl-2-pentanone through the septum and reweigh to within 0.01 mg.

B-1.1.6 Blank Solution — Weigh accurately 0.20 g of the blank into an injection vial. Add 5.0 ml of water and 1.0 ml of the internal standard solution. Heat at 60°C for 10 min and shake vigorously for 10 sec.

B-1.1.7 Samples — Weigh accurately 0.20 g sample into an injection vial. Add 5.0 ml water and add 1.0 ml of the internal standard solution. Heat at 60°C for 10 min and shake vigorously for 10 sec.

B-1.1.8 Calibration Solution — Weigh accurately 0.20 g of the blank into an injection vial. Add 5.0 ml of the water and 1.0 ml of the internal standard solution. Weigh the vial accurately to within 0.01 mg. Inject a known volume of the component of interest through the septum and again reweigh the vial. Heat at 60°C for 10 min and shake vigorously for 10 sec.

B-1.2 Procedure

Place the sample, blank and calibration samples in the sample tray of the head-space gas chromatograph — FID system. Analyze using the following analytical conditions.

B-1.2.1 Column — Fused silica, length 0.8 m, i.d. 0.53 mm, coated with DB-wax, film thickness 1 µm.

B-1.2.2 Coupled with, fused silica, length 30 m, i.d. 0.53 mm, coated with DB-1, film thickness 5 µm.

B-1.2.3

Conditions:

Carrier gas:	Helium
Flow rate:	208 kPa, 5 ml/min
Detector:	FID
Temperatures:	
Injector:	140°C

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Oven conditions:	35°C for 5 min, then 5°C/min to 90°C, then 6 min at 90°
Detector:	300°C

Head space sampler

Sample heating temperature:	60°C
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Sample heating period:	10 min
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Syringe temperature:	70°C
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Transfer temperature:	80°C
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Sample gas injection:	1.0 ml in split mode
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B-1.3 Calculation

$$A \times B \times C/50 = \text{mg component per injection vial}$$

where

A = relative peak area of the component concerned,

B = mg internal standard, and

C = calibration factor.

$$\text{Calibration factor} = D \times 50 / (E \times (F - G))$$

where

D = mg component weighed,

E = mg internal standard,

F = relative peak area of component for the calibration solution, and

G = relative peak area of the same component for the blank solution.

B-2 METHOD II (DETERMINATION CARRIED OUT IN METHANOL)

B-2.1 Reagents

B-2.1.1 Blank, sample with very low solvent content.

B-2.1.2 Internal Standard, 3-methyl-2-pentanone.

B-2.1.3 Methanol

B-2.1.4 Demineralized Water

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B-2.1.5 Internal Standard Solution — Add 50.0 ml methanol to a 50 ml injection vial and seal. Accurately weigh and inject (15 µl 3-methyl-2-pentanone through the septum and reweigh to within 0.01 mg.

B-2.1.6 Blank Solution — Weigh accurately 0.20 g of the blank into an injection vial. Add 5.0 ml of methanol and 1.0 ml of the internal standard solution. Heat at 60°C for 10 min and shake vigorously for 10 sec.

B-2.1.7 Samples — Weigh accurately 0.20 g sample into an injection vial. Add 5.0 ml methanol and add 1.0 ml of the internal standard solution. Heat at 60°C for 10 min and shake vigorously for 10 sec.

B-2.1.8 Calibration Solution — Solution A: Add 50.0 ml methanol to a 50 ml vial and seal. Accurately weigh, to within 0.01 mg, the vial and inject 50 µl of the component of interest through the septum. Reweigh the vial. Mix well.

Weigh into an injection vial, a known amount of blank (0.20 g), add 4.9 ml of methanol and 1.0 ml internal standard solution. Introduce 0.1 ml of Solution A into the injection vial. Mix well and heat at 60°C for 10 min and shake vigorously for 10 sec.

B-2.2 Procedure

Place the sample, blank and calibration samples in the sample tray of the head-space gas chromatograph — FID system. Analyse using the following analytical conditions.

B-2.2.1 Column — Fused silica, length 0.8 m, i.d. 0.53 mm, coated with DB-wax, film thickness 1 µm.

B-2.2.2 Coupled with, fused silica, length 30 m, i.d. 0.53 mm, coated with DB-1, film thickness 5 µm.

B-2.2.3

Conditions:

Carrier gas: Helium
Flow rate: 208 kPa, 5 ml/min
Detector: FID

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Temperatures:

Injector: 140°C
Oven conditions: 35°C for 5 min, then 5°C/min to 90°C, then 6 min at 90°C
Detector: 300°C

Head space sampler

Sample heating temperature: 60°C
Sample heating period: 10 min
Syringe temperature: 70°C
Transfer temperature: 80°C
Sample gas injection: 1.0 ml in split mode

B-2.3 Calculation

$$A \times B \times C/50 = \text{mg component per injection vial}$$

where

A = relative peak area of the component concerned,
 B = mg internal standard, and
 C = calibration factor.

$$\text{Calibration factor} = D \times 50 / (E \times (F - G) \times 10)$$

where

D = mg component weighed,
 E = mg internal standard,
 F = relative peak area of component for the calibration solution, and
 G = relative peak area of the same component for the blank solution.

(FAD 8)